

Proton Beam Irradiation: A Safe Procedure in Postequatorial Extraocular Extension From Uveal Melanoma



IRA SEIBEL, ALINE I. RIECHARDT, KATHARINA ERB-EIGNER, ALEXANDER BÖKER, DINO CORDINI, JENS HEUFELDER, AND ANTONIA M. JOUSSEN

- **PURPOSE:** This study was performed to show long-term outcomes concerning metastasis rates and local recurrence rates after primary proton beam therapy in uveal melanoma with posterior extraocular extension (EOE) with the main focus on optic nerve invasion.
- **DESIGN:** Retrospective case series.
- **METHODS:** All patients treated with primary proton beam therapy for choroidal or ciliary body melanoma with posterior EOE between July 1998 and August 2010 were included. EOE was detected either upon sonography at primary examination or during the surgical application of tantalum clips onto the sclera. Ultrasound was performed in each patient before surgery, and if EOE was detected, a magnetic resonance imaging (MRI) scan was performed to confirm EOE. All patients with tumors exceeding 6 mm in thickness or abutting the optic disc received a 1.5 Tesla MRI scan after clip surgery. To assess EOE during follow-up, either ultrasound examinations or—if initially detected only by MRI—MRI scans were performed during follow-up.
- **RESULTS:** A total of 27 patients underwent primary proton beam therapy. The EOE was separated into 3 growth types: optic nerve infiltration in 10 patients, vortex vein infiltration in 9 patients, and transscleral growth postequatorially in 8 patients. No local recurrences were found during the overall median follow-up of 80 months (11–168 months). Metastasis rates correlated with AJCC stages but not EOE volume.
- **CONCLUSION:** This study shows that posterior EOE can safely be treated by proton beam therapy, even if the optic nerve is infiltrated. MRI enables safe detection of optic nerve invasion. (Am J Ophthalmol 2018;191:49–53. © 2018 Elsevier Inc. All rights reserved.)

BETWEEN 2.3% AND 5.5% OF PATIENTS WITH UVEAL melanoma present with extraocular extension (EOE) at first examination.^{1–3} It has been shown that EOE is more frequently seen in larger tumors, larger anterior tumor extension, diffuse uveal melanoma, monosomy 3, and histopathologic features such as closed vascular loops, epithelioid cell type, and high mitotic rate, resulting in poor survival prognosis.^{4–10} The common assumption is that survival of patients with uveal melanoma and EOE depends not on the respective treatment but on primary tumor characteristics.^{1,8,11,12} Treatment decision depends on EOE location and the extent of volume of EOE. In patients with EOE located posterior to the equator, enucleation is often prioritized.^{3,8} Both brachytherapy and proton beam therapy appear to be safe irradiation modalities of EOE. Thus far, most reported patients presented with EOE located anterior to the equator or were not further classified.^{3,8,11,13–16} Reviewing the literature describes a number of less than 50 patients with primary irradiation for posterior EOE.^{3,8,11,13–15} Nevertheless, local recurrence rates after primary irradiation of posterior EOE are very low with equal rates after plaque radiotherapy or proton beam therapy.^{3,8,11,13} As mentioned before, little information exists about irradiating posterior EOE and particularly optic nerve invasion, which is the main topic of this study.^{3,8,11,13–15} Here, we investigate local outcomes and metastasis rates in patients with posterior EOE according to invaded structures, with a focus on optic nerve invasion.

METHODS

THIS STUDY WAS CONDUCTED IN ACCORDANCE WITH THE Declaration of Helsinki and was approved by the institutional review board of the Charité-Universitätsmedizin Berlin, Germany. All patients with choroidal or ciliary body melanoma and posterior EOE who were treated with primary proton beam therapy between July 1998 and August 2010 were included. Seven patients with anterior EOE were excluded from this study. In preparation for proton beam irradiation, 4 tantalum clips were sutured onto the sclera of patients. Proton beam therapy was administered



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From the Departments of Ophthalmology (I.S., A.I.R., A.B., D.C., J.H., A.M.J.) and Radiology (K.E.-E.), and BerlinProtonen am Helmholtz-Zentrum Berlin für Materialien und Energie (D.C., J.H.), Charité – Universitätsmedizin Berlin, Berlin, Germany.

Inquiries to Ira Seibel, Department of Ophthalmology, Charité – Universitätsmedizin Berlin, Campus Benjamin Franklin, 12203 Berlin, Germany; e-mail: ira.seibel@charite.de

with 68 MeV proton beam with a dose of 60 CGE (cobalt gray equivalent; 1 CGE = 1.1 gray [Gy], taking a radiobiological effectiveness of 1.1 into account) in 4 fractions of 15 CGE per day at the Helmholtz-Zentrum Berlin on 4 consecutive days.¹⁷ Treatment plans for proton beam irradiation were calculated with the treatment planning systems EYEPLAN version 1.2B for patients treated until 2006 and OCTOPUS (OCular TumOr Planning UtilitieS, developed by Deutsches Krebsforschungszentrum, Heidelberg, Germany) version 4.4.9 for patients treated since 2006.^{18,19} If EOE was detected it was included into the “clinical target volume.” The clinical target volume is always extended by a standard 2.5-mm margin to compensate for patient motion, set-up errors, range uncertainties, and lateral dose build-up, as described earlier.²⁰

EOE was detected either upon sonography at primary examination or during the surgical application of tantalum clips onto the sclera. Ultrasound was performed in each patient before surgery, and if EOE was detected, a magnetic resonance imaging (MRI) scan was performed to confirm EOE. The Innovative Imaging I³ System ABD (from Innovative Imaging, Sacramento, California, USA) was used as ultrasound device until 2007, then the V3 from Ellex (Ellex Germany GmbH, Berlin, Germany) until 2011, and from then on the v4 eyecubed from Ellex. All patients with tumors exceeding 6 mm in thickness or abutting the optic disc received a 1.5 Tesla MRI scan with contrast medium after clip surgery. Experienced radiologists specialized in neuroradiology or head and neck radiology evaluated the MRI scans for the occurrence of optic nerve infiltration. Results were always checked by a second specialized radiologist. Therefore, as measurement of the extent of EOE, the volume was calculated during MRI or—if detected during surgery—in detecting the dimensions by a caliper. The volume of growth in vortex veins was calculated with the following formula: Volume = Length of vortex vein invasion × Pi × (radius of vortex vein)².

RESULTS

IN TOTAL, 27 EYES OF 27 PATIENTS WERE TREATED WITH PRIMARY proton beam therapy for posterior EOE. Of these patients (18 male and 9 female), 5 patients showed ciliary body involvement. The EOE was separated into 3 growth types: optic nerve infiltration in 10 patients, vortex vein infiltration in 9 patients, and transscleral growth postequatorially in 8 patients. The Table shows tumor characteristics, AJCC categories, AJCC stages, EOE characteristics, and follow-up-time according to EOE growth type. Optic nerve infiltration was initially detected by ultrasound in 2 patients and by MRI in 8 patients. Vortex vein infiltration was discovered only during clip surgery in all 9 patients. Posterior transscleral EOE was initially detected by ultrasound in 7 patients and during clip surgery in 1 patient.

To assess EOE during follow-up, either ultrasound examinations or—if initially detected only by MRI—MRI scans were regularly performed during follow-up.

No local recurrences were found during the overall median follow-up of 80 months (11–168 months) (mean follow-up 73 months, 95% confidence interval [CI]: 57–90 months). In total, 3 eyes underwent secondary enucleation owing to unresponsive neovascular glaucoma. One of these had optic nerve infiltration EOE, 1 posterior transscleral EOE, and 1 vortex vein invasion. In all 3 enucleated eyes, EOE was histologically confirmed (Supplemental Figures 1 and 2; Supplemental Material available at AJO.com).

Median follow-up was 98 months (30–168 months) in patients with optic nerve infiltration, 55 months (11–140 months) in patients with posterior transscleral EOE, and 40 months (14–135 months) in patients with vortex vein invasion. Univariate Cox regression analysis comparing optic nerve infiltration and posterior transscleral growth showed no statistically significant difference in the development of metastases. Univariate Cox regression analysis comparing vortex vein infiltration with optic nerve infiltration showed a 2.3 times higher hazard developing metastasis in patients with vortex vein infiltration (95% CI: 1.056–5.130, $P = .036$) (Figure). Univariate Cox regression analysis for largest basal diameter (LBD) showed a 1.8 times higher hazard (95% CI: 1.2–2.6, $P = .008$). As patients presenting with vortex vein infiltration had statistically significantly larger basal diameter and thicker tumors, with $P = .0001$ (Fisher exact test) a bivariate Cox regression analysis was performed. A bivariate model that included the type of EOE and LBD showed that LBD was a predictor of metastatic risk with a hazard ratio of 2.0 (95% CI: 1.1–3.8, $P = .02$), while the type of EOE was not independently associated with metastasis (hazard ratio 0.6; 95% CI: 0.2–1.9, $P = .38$). No statistically significant difference was found in terms of EOE volume.

DISCUSSION

NO LOCAL RECURRENCE WAS FOUND IN THIS STUDY AFTER A median follow-up period of 80 months. This is consistent with the existing literature for posterior EOE.¹⁵ EOE was also found not to be a risk factor for the loss of local tumor control in 119 patients treated by Egger and associates with proton therapy. Unfortunately, EOE has not been further classified into anterior or posterior growth.¹³ Only Gragoudas and associates described extraocular extension as a borderline factor in univariate analysis for local failure of proton beam therapy.¹⁴ Local recurrence rates after irradiation of posterior EOE are equally low. Gündüz and associates treated 3 of 3 patients with posterior EOE⁸ and Augsburger and associates 7 of 8 patients successfully with brachytherapy.¹¹ Proton beam therapy has the main

TABLE. Tumor and Extraocular Extension Features, AJCC Categories, and Stages, in Total and According to Groups of Extraocular Extension

EOE Growth Type	Optic Nerve Infiltration	Vortex Vein Infiltration	Transscleral Posterior Growth	Total
Number of patients with EOE	10	9	8	27
Median (range) follow-up, months	98 (30–168)	40 (14–135)	55 (11–140)	80 (11–168)
Median largest diameter of EOE in mm	3 (2.5–4)	1.5 (0.5–3)	3.25 (1.5–11)	2.5 (0.5–11)
Median (range) EOE volume in mm ³	12.8 (8–288.0)	6.3 (0.8–15.7)	19.5 (3.5–476.0)	8.8 (1.0–476.0)
AJCC category	T1c: 1 T2c: 6 T3c: 3	T3c: 3 T4c: 1 T4d: 5	T1c: 1 T2c: 1 T3c: 3 T4d: 1 T4e: 2	T1c: 2 T2c: 7 T3c: 9 T4c: 1 T4d: 6 T4e: 2
AJCC stages	IIA: 1 IIIA: 9	IIIA: 3 IIIB: 1 IIIC: 5	IIA: 1 IIIA: 4 IIIC: 3	IIA: 2 IIIA: 16 IIIB: 1 IIIC: 8
Median (range) tumor thickness in mm	4.4 (2.9–11.6)	10.5 (8.2–13.1)	6.7 (3.6–11.8)	8 (2.9–13.1)
Median (range) largest basal diameter in mm	11.0 (8.4–17.5)	18.8 (13.2–20.0)	14.0 (8.7–21.4)	15.5 (8.4–22)
Median (range) distance to fovea in mm	1.6 (0.0–3.0)	3.2 (0.0–13.0)	3.1 (0.0–7.0)	3.1 (0.0–15.3)
Median (range) distance to optic disc in mm	0.0 (0.0–0.0)	4.0 (1.4–11.6)	3.1 (0.0–5.0)	3.3 (0.0–15.3)
Median (range) dose to optic disc in CGE	60.0 (40.0–60.0)	6.2 (0.0–59.4)	2.7 (0.0–60.0)	6.2 (0.0–60.0)
Median (range) irradiated optic nerve length in mm	3.4 (1.8–4.3)	0.2 (0.0–2.3)	0.25 (0.0–3.9)	0.5 (0.0–4.3)
CGE = cobalt gray equivalent; EOE = extraocular extension.				

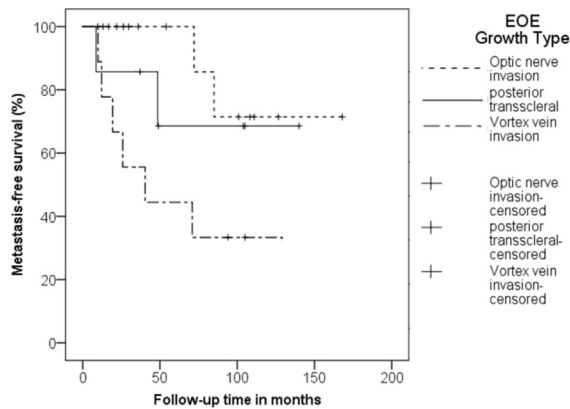
advantage that treatment planning can be adjusted. If EOE is detected during preparation for the irradiation, the clinical target volume can be expanded according to the size of EOE. Since each irradiation plan for proton beam treatment is created individually for every single patient, it can be adapted to the extent of the EOE. This expansion of the target volume to include the extraocular growth volume is done either by means of images or, in the case of smaller-volume EOE such as vortex vein infiltration, by exact localization and description of the surgeon. By measuring the extraocular growth to the limbus and to the clips placed, it is possible to precisely determine the position and make it comprehensible for the medical physicists who construct the radiation model ([Supplemental Figures 3–5](#); Supplemental Material available at [AJO.com](#)).

Thus, even after accidental opening of the sub-Tenon space during the operation, the entire area can be irradiated with a sufficient safety margin. EOE along the vortex veins was associated with a higher risk for developing metastasis. These patients showed worse AJCC prognostic stages owing to statistically significantly larger basal diameters and thicker tumors than patients developing other types of EOE—both features representing poorer prognosis.¹⁵ Furthermore, it was previously shown that the diameter is a risk factor for EOE and strongly associated with vortex vein invasion.^{10,15} In all patients with EOE, optic nerve

infiltration with a metastasis-free survival of 142.4 months related to a superior prognosis in our cohort. In accordance with the literature, this may be explained by better AJCC prognostic stages, a smaller tumor thickness, and a smaller basal diameter in this group.^{1,2,12} These results confirm once again that the survival of patients with uveal melanoma and EOE is dependent on the primary tumor characteristics.^{1,5,14,15} EOE with a diameter exceeding 5 mm was considered an additional risk factor for metastasis.^{1,21}

In this study only 2 patients with EOE with a diameter of more than 5 mm were irradiated. Therefore no statement about a statistical difference according to AJCC categories can be made. Of both patients, 1 died owing to metastases after 11 months; the second patient is metastasis-free with a follow-up period of 104 months.

Detection of posterior EOE can be challenging. With a small EOE volume, the ultrasound examination showed a low sensitivity in the present study, which is consistent with the results of Bercher and associates,¹⁵ whereas Gündüz and associates have always found the EOE by means of ultrasound sonography.⁸ Small EOE volume and an ultrasound device that was older at the time of data collection certainly made diagnosis by ultrasound more difficult. Vortex vein infiltration was only seen intraoperatively. The surgeon should be aware of the risk of



	After year	0	1	2	3	4	5	6	7	8	9	10
Optic nerve invasion	Numbers at risk	10	10	10	10	10	10	10	9	8	8	8
Posterior transscleral	Numbers at risk	7	7	6	6	6	6	5	5	5	5	5
Vortex vein invasion	Numbers at risk	10	9	8	6	4	4	3	3	3	3	3

FIGURE. Metastasis-free survival according to groups of extraocular extension (EOE) from uveal melanoma after primary proton beam therapy. Kaplan-Meier survival analysis calculated a metastasis-free survival after 5 years of 71% in patients with optic nerve infiltration, 59% in patients with posterior transscleral growth, and 44% in patients with vortex vein invasion. Log-rank test showed that the metastasis-free survival differed statistically significantly ($P = .04$). Kaplan-Meier estimates calculated an overall metastasis-free survival of 106.0 months (95% confidence interval [CI]: 79.6–132.5 months). The metastasis-free survival according to groups was calculated to 142.4 months (95% CI: 112.4–172.5 months) in patients with optic nerve invasion, 105.5 months (95% CI: 65.3–145.7 months) in posterior transscleral EOE, and 62.9 months (95% CI: 27.7–98.0 months) in vortex vein invasion.

extraocular growth—especially with large basal diameters—and should therefore carefully prepare backwards to avoid opening the sub-Tenon space. Since this type of posterior extraocular growth had the smallest volume, any extraocular growth was hardly visible in postoperative MRI and ultrasound. For this reason, a tumor with extraocular growth along the vortex veins should be considered to be well controlled as long as no new growth in ultrasound or MRI was visible during follow-up examinations.

The incidence of optic nerve infiltration is about 5.7% and is limited to choroidal and ciliary body melanoma.²² In our study the EOE was usually found by MRI, which makes this extended imaging recommendable. Nonetheless, patients with EOE may have been unnoticed. An MRI was only performed if the tumor was thicker than 6 mm or was found to be funduscopically juxtapapillary. Lindegaaard and associates reported optic nerve invasion in up to 34% of patients without funduscopically visible juxtapapillary growth.²² Based on their multivariate analysis, not only patients with a juxtapapillary melanoma but also those with an intraocular pressure of more than 24 mm Hg should probably be examined using MRI.²² We therefore definitely recommend MRI scans be conducted in all patients with tumors touching the optic disc.

This study has several limitations. Even if the diagnosis of EOE along the optic nerve was always performed by 2 specialized radiologists, it was not proven histologically. Although not relevant for treatment, no conclusive statement can be made about the exact origin of the EOE without histology or an increase in resolution in future imaging (eg, an invasion of the posterior ciliary arteries surrounding the optic nerve could be confused with an invasion of the optic nerve). Still, MRI is very useful in diagnosing choroidal melanoma. By using diffusion-weighted MRI, its accuracy has been significantly increased because structures can be differentiated on the basis of their cellular density.²³ As a consequence, diffusion-weighted imaging has become a mandatory part of the ocular MRI protocol in our oncology service since 2012.

The main strength of this study is that this is, to our knowledge, the largest series that treated optic nerve invasion EOE with proton beam therapy. Our results show that proton beam therapy allows safe irradiation of posterior EOE, including optic nerve infiltration. Furthermore, MRI enables detection of optic nerve invasion that may occur even in small melanoma. Therefore we strongly recommend an extended imaging with MRI for all uveal melanoma touching the optic disc (Supplemental Figures 6 and 7; Supplemental Material available at AJO.com).

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